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Stereoselective Synthesis of Polysubstituted Olefins via the Formation of 1,1-Bimetalloalkenes

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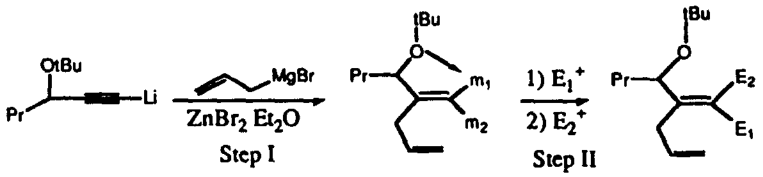
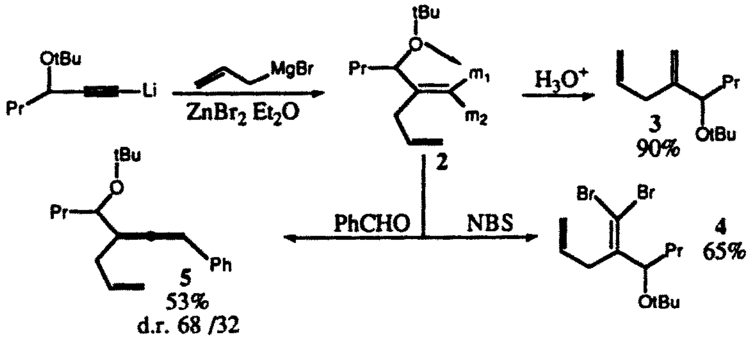
Abstract: The stereoselective reaction of 1,1-bimetalloalkenes with different electrophiles allows the synthesis of the corresponding polysubstituted olefins

1,1-Bimetalloalkenes are versatile reagents and the use of a variety of metals including aluminium and titanium[1], aluminium and zirconium[1], zinc and zirconium[2] has been investigated as ketene-transfer reagents. Other 1,1-bimetalloalkenes[3] containing zinc and boron[4], copper and boron[4], lithium and boron[5] and finally boron and zirconium[6] were also investigated as a source of stereodefined polysubstituted C-C double bonds. In the course of our study on the formation of organogembismetallics via an allylmetallation of metallated unsaturated systems[7], we now report that the allylzincation of alkynyl metals (step I, Scheme 1) opens a new route to stereodefined tri- and tetrasubstituted double bonds[4,5,6,8]. Our strategy for effecting such reaction with high stereoselectivity requires a substrate bearing a lewis basic functional group[9] which serves to differentiate the reactivity of the two metals of the 1,1-bimetalloalkene toward electrophiles (step II, Scheme 1).



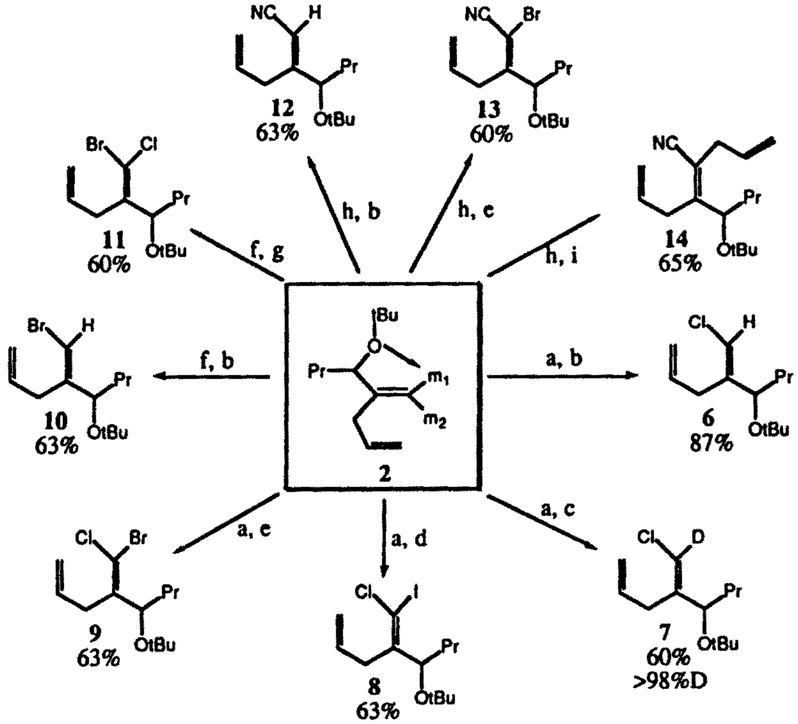
Scheme 1

The first step, originally described by Gaudemar using THF at reflux[10], has been performed under very mild conditions in a less basic solvent such as Et_2O (-10°C, 30 mn, 90% yield) and the presence of the 1,1-dimetallic species has been proved by reaction of 2 with electrophiles.



Scheme 2

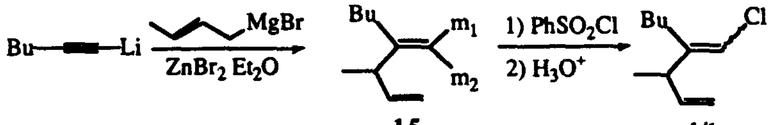
The second step of the Scheme 1, namely the stereoselective reaction of the organogembismetallic with different electrophiles, allowed the synthesis of the corresponding tri and tetrasubstituted olefins (Scheme 3). Thus, addition of phenylsulfonylhalides to the organogembismetallic[11] leads to the corresponding vinylic carbenoid. The hydrolysis (or deuterolysis) of the latter gives the corresponding unsaturated vinyl halide in good isolated yield.



Reaction conditions : a) 4 eq. TosCl, -20°C, 1.5h b) H_3O^+, -20°C. c) DCl, -20°C to r.t. d) I_2 THF, -30°C. e) NBS, CH_2Cl_2, -30°C. f) 2 eq pMeTosBr, Et_2O, -50°C, 15 mn. g) NCS, CH_2Cl_2, -50°C to -20°C. h) 2 eq TosCN, Et_2O, -50°C to r.t. then 2h. at r.t. i) $Me_2CuCNLi_2$ THF, -20°C, 15mn then allyl bromide, $P(OEt)_3$, -40°C, overnight

Scheme 3

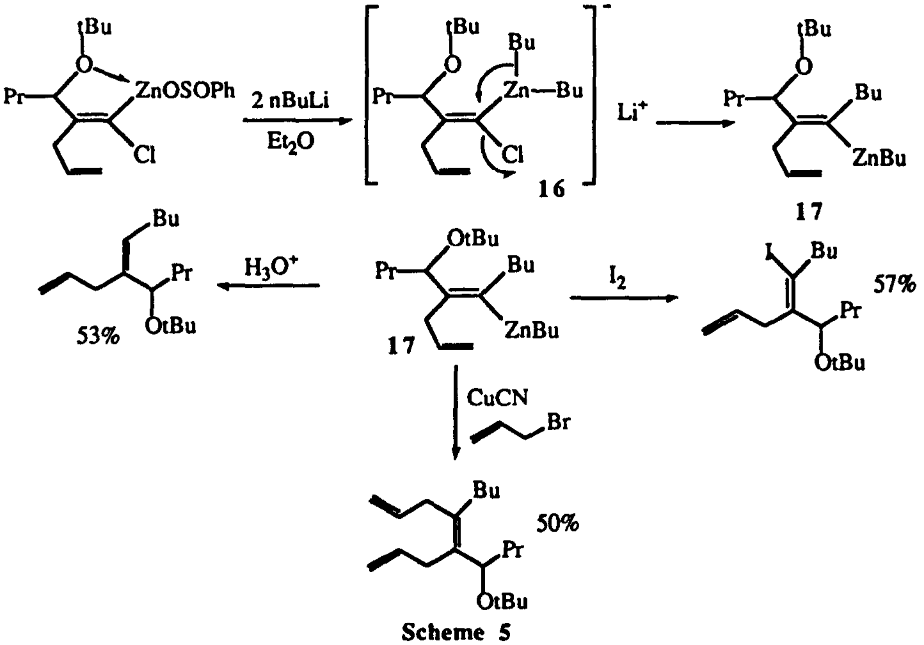
In all cases examined, products are obtained as a unique stereoisomer (determined by N.O.E. effect) which reflects the chelation of the terbutoxy group toward m_1 of **2**. The chelation is absolutely necessary to discriminate the reactivity of the two metals toward the first electrophile. This was checked by the reaction with a substrate without internal chelation : **15**[12] leads to the two geometrical isomers in equal amounts (Scheme 4).



Scheme 4

Coming back to our initial concern, the remaining metal (m_2) can also react with a stronger halogenating agent (Scheme 3) to give the polyhalogenated vinylic compound as a unique isomer. This reaction is of synthetic value for the obtention, at will, of the two geometrical isomers (**9** versus **11**) starting from the same propargylic ether. Interestingly, the addition of p-toluenesulfonyl cyanide[13] to **2** leads also to a unique metallated vinyl cyanide which can be hydrolyzed, bromolyzed or transmetallated to an organocopper derivative and allylated with 65% isolated yield, (see **12**, **13**, **14**).

This strategy allows, in a one-pot sequence, the preparation of a stereodefined zinc chorocarbenoid which is configurationnally stable at low temperature. The dropwise addition of nBuLi (2 equivalents) to this mixture at -78°C leads to the formation of the zincate carbenoid **16** which undergoes a smooth intramolecular nucleophilic substitution reaction[14] to give the allylated vinylic organozinc derivative **17**, characterized by hydrolysis, iodolysis and allylation with a 92/8 stereochemical purity.



Scheme 5

Thus, an efficient and straightforward route to tri- and tetrasubstituted olefins has been developed, starting from a very common propargylic ether.

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